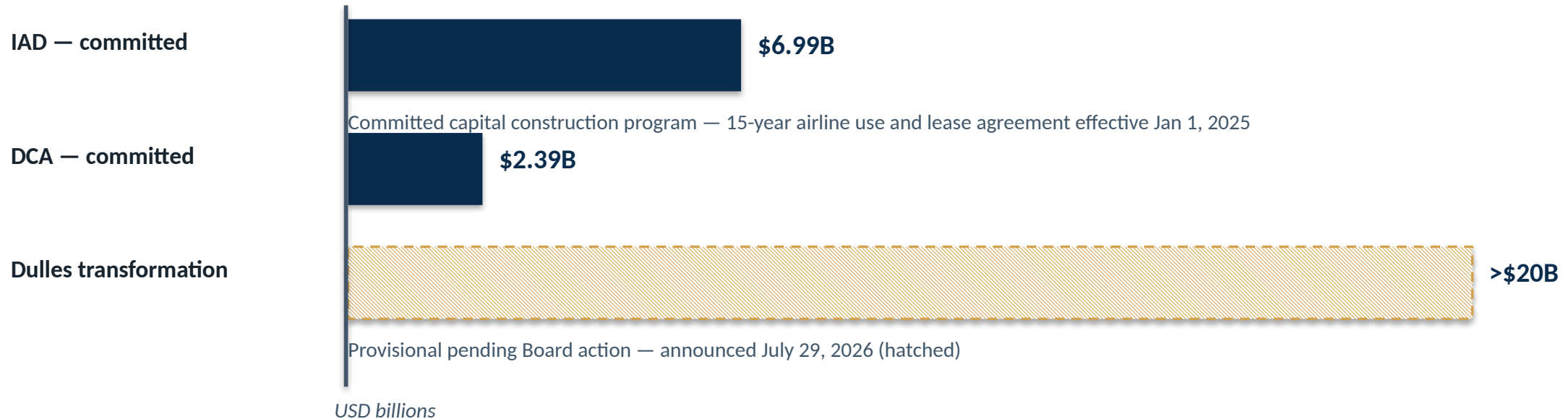

The loud terminal is the unintelligible one

A communications-integrity standard for MWAA terminals — argued as safety engineering, delivered through the Design Manual.

Technical read-ahead · Metropolitan Washington Airports Authority · August 2026

Roughly \$9B committed, \$20B+ provisional — all of it passes through the Design Manual



Every square foot of that work passes through the MWAA Design Manual — “a mandatory guide with the force of law on the airport property.”

After January 29, 2025, no board signs a policy paraphrasable as 'fewer safety broadcasts'

THE CONSTRAINT

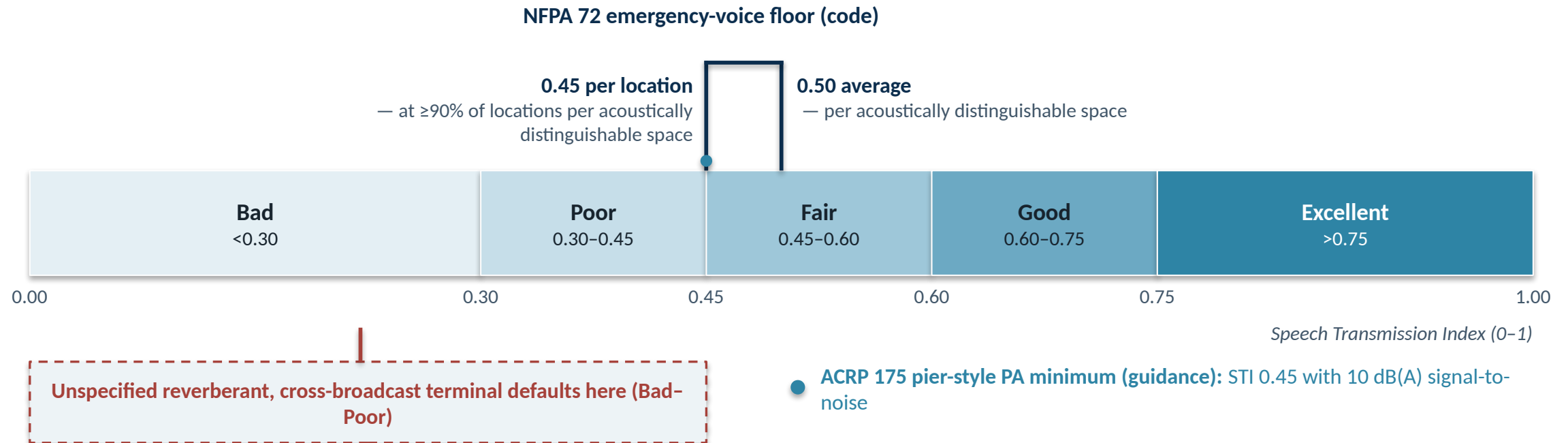
The NTSB's final report on the January 29, 2025 midair collision over the Potomac — 67 fatalities — resets public tolerance for any policy paraphrasable as “reduce safety announcements.”

THE INVERSION

The standard is defensible only as the opposite of subtraction: a floor that raises the probability the announcement that matters — the gate change, the shelter-in-place, the IROPS diversion — reaches the passenger who needs it.

One ruler decides the argument: the Speech Transmission Index

A terminal fails these floors by being loud, reverberant, and cross-broadcast — precisely the unspecified default.



Code: NFPA 72 (adopted edition governs) · Guidance: ACRP Research Report 175 (TRB, 2017) · Measurement method: IEC 60268-16:2020

The Joint Commission did not silence ICU alarms — it made hospitals govern them

The problem, documented

Jan 1, 2014 — NPSG.06.01.01 phase one

Jan 1, 2016 — phase two



One academic medical center logged 74,535 alarms across eight units in one week (illustrative magnitude). Clinicians had stopped hearing them.

Every accredited hospital must inventory and prioritize its clinical alarms.

Governance policies, accountability, and proof on the record — not silence.

The pattern maps directly to airports: inventory, rank, govern, log.

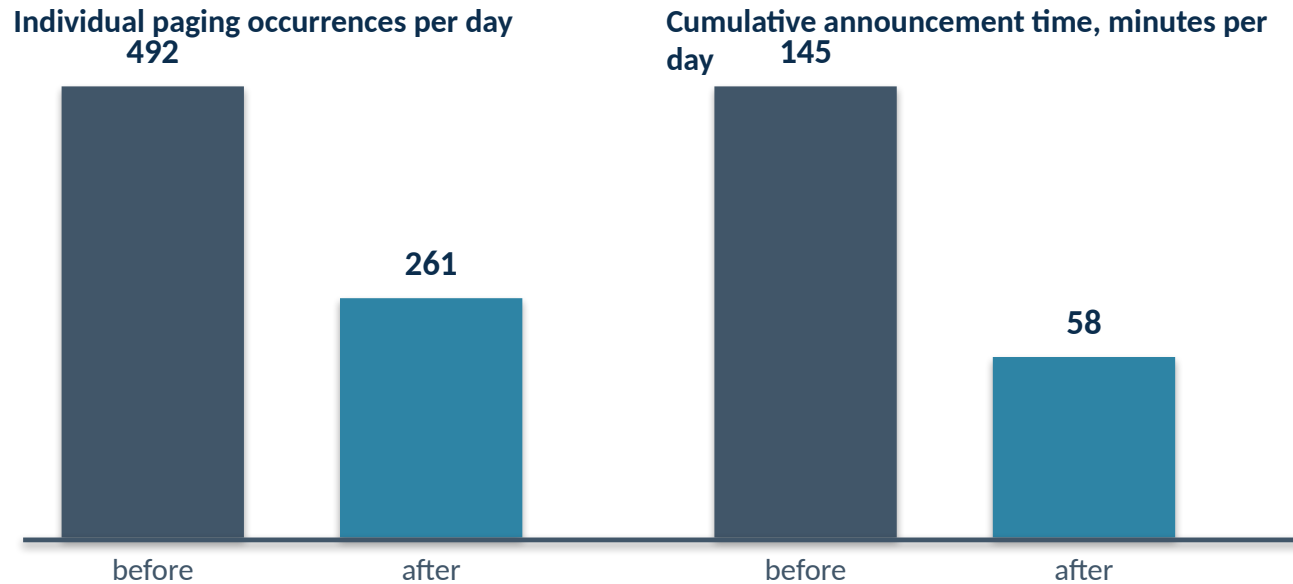
The 'silent airport' peer set is a category error — none published measurement

Airport	Announcement policy (operator-described)	Structural profile vs. IAD/DCA	Published intelligibility or missed-boarding data
London City (LCY)	Silent policy since 2008; no flight or gate announcements since Aug 18, 2016, except essential and emergency	Small, point-to-point, business-traveler-dominant	None identified in this research pass
Helsinki-Vantaa (HEL)	Silent regime since 2015; gate-area and emergency overrides retained	Small transfer airport, high English proficiency	
Copenhagen (CPH)	Gate and boarding information moved to screens and app	Point-to-point-weighted Northern European hub	
Amsterdam Schiphol (AMS)	Operator-described “Silent Airport”; information on screens and airport app	Large transfer hub — the closest structural analogue; rests on screen density and app-carrying passengers	

Policy analogues, not design precedents.

SFO shows the operating model — and what subtraction without engineering costs

SFO-reported; methodology not published



THE SAME RECORD

Documented passenger complaints and missed-flight accounts attributed to the reduced-announcement practice, in contemporaneous reporting.

Announcement reduction without intelligibility engineering and channel redundancy is subtraction, not design.

The people who most need to hear are the legal floor, not an edge case

28 CFR 35.160 — ADA Title II Effective communication with people with disabilities; auxiliary aids and services.

2010 ADA Standards §219 / §706 Assistive-listening systems in each assembly area where audible communication is integral to use.

14 CFR Part 382 Prompt access to the same gate information for passengers needing visual or hearing assistance.

ACRP Research Report 239 Digital-first substitution is partial, not sufficient, for travelers with disabilities and older adults.

guidance

A standard that mandates redundant visual, digital, staffed, and hearing-loop channels binds MWAA harder than current practice, not softer.

Reducing broadcast without those redundancies compounds exclusion.

A two-tier standard — and the paragraph the silent-airport literature omits

MANDATORY AT BOTH AIRPORTS

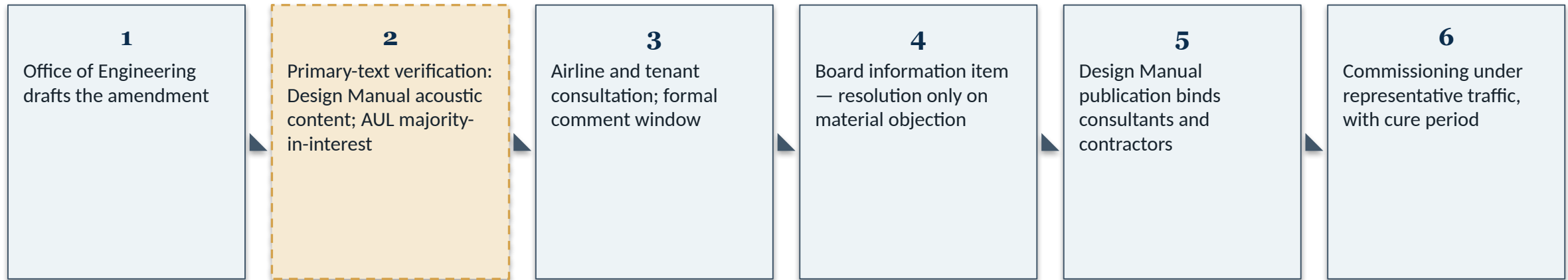
- Acoustically-distinguishable-space (ADS) map as a design deliverable
- Intelligibility floors per NFPA 72 and ACRP 175
- Measurement per IEC 60268-16:2020, commissioned under representative traffic, with cure period
- Accessibility floors: 28 CFR 35.160, 2010 ADA Standards, 14 CFR Part 382
- Hearing conservation per 29 CFR 1910.95
- Audit-logged announcement governance
- Concessionaire TVs and gate-lounge audio kept in scope

CONTEXT-DEPENDENT BY TERMINAL

- Tier levels for PA zoning and announcement density
- Quiet and low-stimulation space scope
- Enforcement mechanics by lease type

IROPS OVERRIDE — during declared irregular operations, the Airport Duty Manager holds named authority to exceed announcement-density templates; every deviation is logged and reconciled afterward.

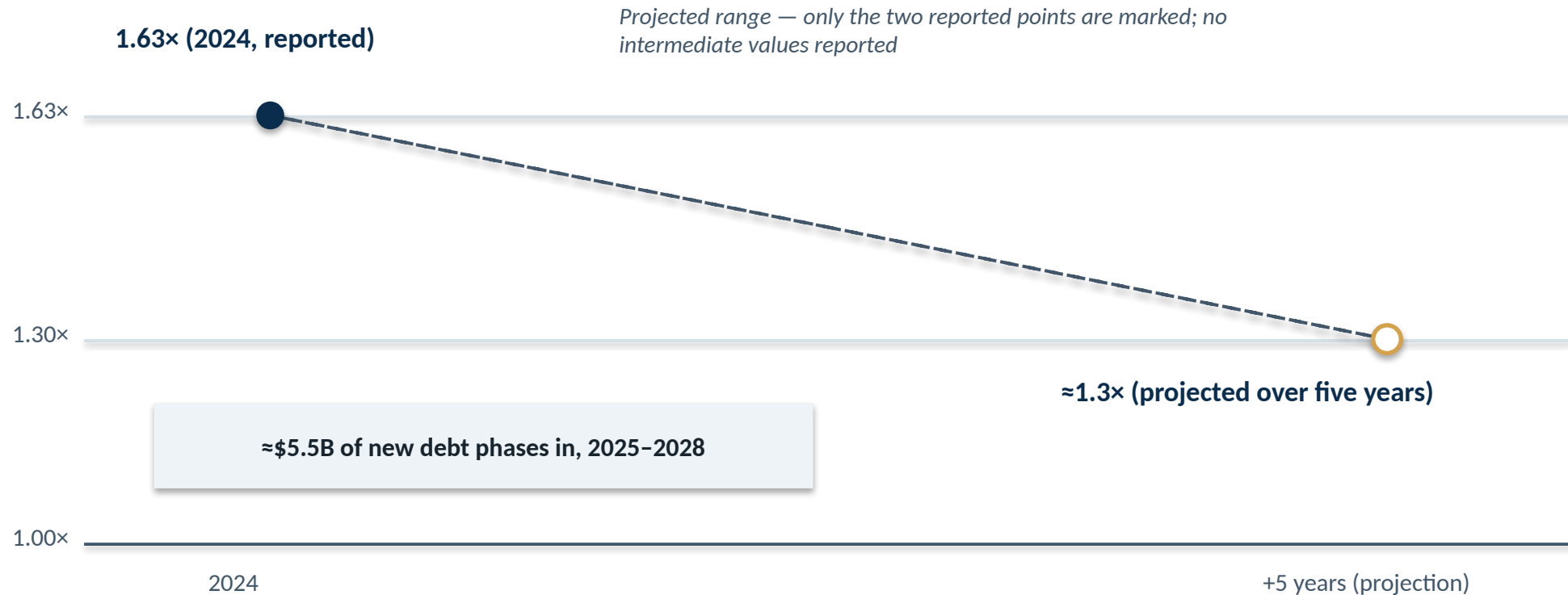
The ask: write the paragraph this capital cycle



First 90 days: commission the Concourse E intelligibility and ambient baseline at the fall 2026 opening. Acceptance floor: $STI \geq 0.45$ at $\geq 90\%$ of locations per acoustically distinguishable space, average ≥ 0.50 , measured per IEC 60268-16:2020 under representative traffic.

Read-ahead: the coverage window is why the amendment targets work still in programming

Net-revenue debt-service coverage (x)



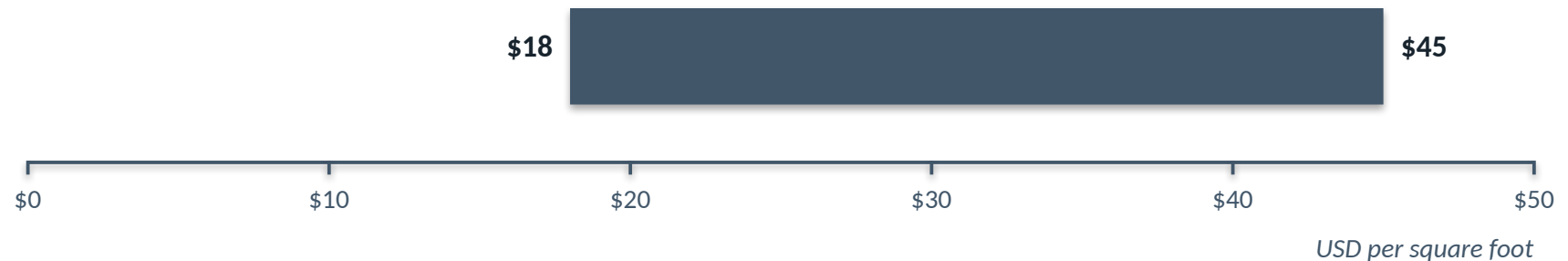
Added specification cost inside programs already in design cuts against coverage; specification at programming stage does not.

Read-ahead: specify at design time — the retrofit premium at commercial benchmarks

New construction — acoustic ceiling treatment



Retrofit — acoustic panel treatment



Non-airport commercial benchmarks, medium confidence — airside premiums would raise both ranges.

The ranges point one direction: specify at design time.

Read-ahead: one standard, two very different anchor-tenant problems

IAD — INTERNATIONAL-CONNECTION HUB

- 29.01M passengers in 2025; 10.53M international
- Fastest 2025 growth among the 50 largest US airports
- United: ≈67% of passenger traffic; 50 jetbridge-equipped gates in Concourses C and D

DCA — SLOT-CONTROLLED DOMESTIC O&D

- American: ≈53% market share (end-2024)
- 2025 traffic declined — the January collision, softened government-related travel, a weeks-long federal shutdown
- Podium PA effectively airline-owned at preferential-use gates

Airline briefing reported by Cranky Flier (Aug 4, 2026): **fully loaded IAD transformation cost per enplanement ≈\$90.64 vs. \$12.88 today** — analyst-reported airline positioning, not an MWAA figure.

Read-ahead: half the soundscape is tenant-owned — governance is the technology

HOLDROOM — conceptual, not a floorplan

MWAA-controlled overhead PA zones

Governed by the Design Manual today; the amendment adds intelligibility floors, ADS mapping, and audit-logged governance.

Airline podium PA (preferential-use gate)

Amendment written to reach it — contingent on AUL majority-in-interest verification.

Concessionaire TVs and audio

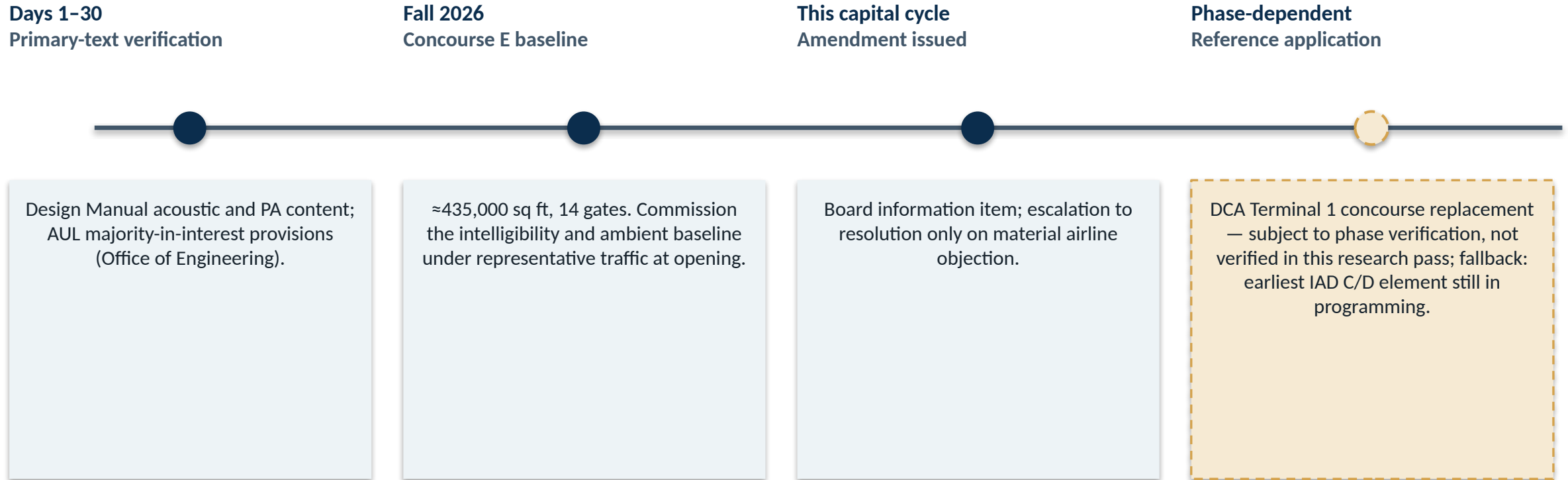
Lease terms; ambient thresholds set in the amendment.

AUDITABLE GOVERNANCE THE AMENDMENT CAN SPECIFY

- Centralized management
- Replay and audit logs of every broadcast
- Elimination of manual-initiation errors
- Templated multi-language delivery

Enforcement: annual acoustic inspections keyed to lease renewal · cure notices · lease-management escalation for pattern noncompliance.

Read-ahead: Concourse E carries the first commissioned baseline this fall



The Concourse E baseline becomes the number every future MWAA facility is measured against.

Read-ahead: an empty-building reading is not a commissioning result

1 Empty-building STI baseline

Measured per IEC 60268-16:2020 — necessary, but not a commissioning result.

2 Representative-traffic verification

Required because the standard's own scope (§7.13, §8.9.3) does not cover fluctuating background noise — and terminals are the paradigm fluctuating-noise environment.

3 Cure period and re-measurement

Verification-with-cure-period, not an occupancy condition — keeps acceptance off the occupancy critical path.

Related floors: 29 CFR 1910.95 — hearing-conservation program at 85 dB(A) eight-hour TWA · ANSI/ASA S12.60 — 35 dB(A) core-learning-space background, the reference benchmark for MWAA-defined low-stimulation zones only (reference, not terminal ambient).

Read-ahead: 'the floors already exist' is half-right — here is the marginal content

WHAT EXISTING FLOORS REQUIRE

NFPA 72 — emergency voice intelligibility only; no ADS mapping as a general deliverable, no routine-PA governance

2010 ADA Standards — receiver counts, not intelligibility floors for daily announcements

ACRP 175 — guidance, not code

WHAT ONLY THE AMENDMENT ADDS

- ADS map at each design milestone
- Representative-traffic commissioning
- Audit-logged governance of routine PA
- Channel-redundancy floors
- IROPS override
- Tenant-audio boundaries

The serious objection: premature specification before program-brief maturity drives change orders, and hard commissioning gates on the critical path in this coverage window would be a bond-rating conversation. Answered by cure-period commissioning and milestone — not occupancy — deliverables.

Read-ahead: what this research pass did not verify

Open verification item	Owner	Closure route
Current Design Manual acoustic, PA-zoning, and STI content	Office of Engineering	Read Volume 2 and applicable specification sections (public, mwaa.com)
2025 AUL majority-in-interest thresholds and cost-recovery categorization	Office of Engineering with airline relations	Primary agreement text
DCA Terminal 1 concourse replacement project phase	Capital program office	Capital program status reporting
SFO Quiet Airport measurement methodology	Research follow-up	SFO Airport Commission minutes or program report
Rating-agency primary reports behind the Bond Buyer coverage figures	Finance	Moody's, Fitch, and S&P MWAA reports

Every figure in this packet carries a documented source; the full source appendix is in the companion report. Nothing on this page is asserted as verified.

Sources: open items per the report's evidence review. Closing them is the first 90 days of assigned work.